PA194HAZ: a 194-state PA consistency machine

Complete corpus comparison, a verified composition,
and reproducible reduction searches

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Abstract

We give a complete binary single-tape machine, PA194HAZ, with 194 working states and an ordinary mathematical proof, formalizable in PA using the supplied development's unchanged calibration and arithmetic-block lemmas, that its blank nonhalting is equivalent to Con(PA). The machine changes five transitions of the primary PA194 table. It combines the supplied MPH cached-result protocol, the A/D physical register exchange, and a known-zero AX closure. The four-transition companion PA194HA is also supplied. Each constituent idea has a prior instance in the supplied corpus; the contribution is the particular composition, its explicit command-level proof, its source invariant, and its verified physical realization.

The audit covers all 41 placed PA research packets and all five incoming archives at a pinned revision. Their 3,597 recognized table occurrences contain 666 byte forms. The 664 forms with a spanning control root give 476 action-graph classes, including 28 classes at 194 states. Both new machines differ from every recorded 194-state class even under control renaming, arbitrary spanning-root choice, tape reflection, and global symbol complementation. Auxiliary and rejected tables are included.

Executed Python and JavaScript checks agree on all 4,751 outcomes of HAZ's 3,650-site source and its exact 58,122-step blank initialization. A separate finite control-graph checker validates the four-site invariant. No sub-194 machine was obtained, and no lower bound or new proof-assistant endpoint is claimed. Complete tables, source, proofs, corpus data, negative search receipts, and offline reproduction programs accompany this report.

Contents

1 The result and its precise scope **3**
1.1 Five exact edits . . . 3
1.2 Contribution and proof status . . . 3

2 Comparison of the complete supplied corpus **4**
2.1 Pinned scope and reproducible census . . . 4
2.2 How graph equivalence was tested . . . 5
2.3 The prior 194-state designs . . . 6
2.4 Larger designs and why reductions do not simply add . . . 7

3 Numerical calculus and the hybrid command interface **7**
3.1 Commands . . . 8

4 A total Pop operation and complete proof compilation **8**

5 Soundness and PA formalizability **10**

6 Source construction and physical verification **11**
6.1 The HA source and register layout . . . 11
6.2 The four-site zero-register certificate . . . 11
6.3 Universal source contracts and exact initialization . . . 12

7 Attempts to get below 194 states **13**
7.1 Bounded physical quotients . . . 13
7.2 Transfer from ZFC and concise iPRA . . . 13

8 Reproduction and contents **14**

A The 28 prior 194-state graph classes **15**

B All 41 placed research packets **16**

C Complete HAZ transition table **18**

1 The result and its precise scope

The machine uses alphabet $\{0, 1\}$ on a two-way infinite tape, initially zero everywhere, with the head at cell zero. Every transition writes one bit and moves one cell left or right. There are 194 working states; the halt label is separate. In the dense table the start is zero, working states are $0, \ldots, 193$, and halt is 194. The native start is 74 and the native halt is 254. All 194 working controls are graph-reachable from the start; this is a graph property, not a claim that the blank run visits them all.

The claimed arithmetic result is

$$\mathrm{PA} \vdash (\neg\,\mathrm{Halt}(\mathrm{PA194HAZ}, \mathrm{blank}) \leftrightarrow \mathrm{Con}(\mathrm{PA})),$$

where Halt is the usual existential finite-run predicate and Con(PA) is the absence of a PA proof of $0 = 1$. The proof is given below by composing finite syntactic proof transformations and source-to-machine simulation. The corresponding statement also holds for PA194HA.

1.1 Five exact edits

Take the primary native table of [2] and apply the following five replacements. No other transition changes.

Native state	Read	Original action	HAZ action	Purpose
17	1	$0R32$	$0R48$	A/D register placement
144	1	$0R48$	$0R32$	A/D register placement
161	1	$0R32$	$0R48$	A/D register placement
164	1	$1R98$	$1R130$	MP returns to cache
2066	1	$0L302$	$1R21$	Known-zero AX closure

PA194HA applies the first four edits. HAZ additionally implements the AX return jump by a decrement of logical register X, whose value is proved zero at that point. A full graph comparison distinguishes HA, HAZ, and every one of the 28 prior classes from each other.

Dense labels are assigned by breadth-first traversal from native 74, visiting read zero before read one; halt is assigned 194. The dense HAZ table is printed in Appendix C. Each file has 388 rows in the format `state read write direction next`, with LF line endings. The exact SHA-256 identities are:

HAZ native: `5ffbc263413612caa71b85895a92860797d90f5c77315119134b6e30fd566f09`

HAZ dense: `f9f98218e2f949b28f0429f2bbab17babe0f4948c2e38cb786a5e3b239101981`

HA native: `503e27b6b619eb04d567d94e47b3bfe667eb457bb921eab3efc4d5d4771036bc`

HA dense: `dd8b4b0a6419dde553bf840175c5e682b7b1f574f5c152278fb6dfbf2916d7e3`

1.2 Contribution and proof status

The initial one-edit candidate, called H during this investigation, proved to be byte-for-byte identical to `research/physical/inputs/MPH.native.tm` in the N archive. It is an existing design and is credited as such. Neither the cached-result MP idea, total Pop, the A/D exchange, nor replacing a known-zero jump is claimed as new. HA and HAZ supply explicit combinations absent from all materialized literal tables in the requested corpus. HAZ changes the successful-MP

store semantics relative to primary PA194, the physical representation of A and D, and the primitive chosen for the AX return. Its novelty claim is limited to that concrete construction and proof, not to a new logical calculus.

The proof has four layers: command soundness and all-store completeness; the unchanged numerical-calculus/PA refutation translations; universal source-instruction refinement with inherited arithmetic-block contracts; and exact complete-tape initialization. Python and JavaScript execute the finite checks. The mathematical lemmas and the claims of PA formalizability are ordinary proofs. This work does not supply a new end-to-end Lean or Rocq proof term. The repository's separate machine-specific iPRA endpoint remains a distinct formalization task [1].

2 Comparison of the complete supplied corpus

2.1 Pinned scope and reproducible census

All retrieval and comparison use repository revision

`1e407ae841e4666f932777fad977ba784ebbcfd5`.

The inventory includes every direct `.tm` file in the 41 packet directories under `Computability/BusyBeaver/Research/PA/docs`, and every complete five-column numeric binary table found in `.tm`, `.txt`, `.csv`, or `.tsv` members of the five incoming archives. All five archives were decoded. Reports, source descriptions, proofs and relevant search families were also inspected [6]. Arbitrary code encodings and every unmaterialized parameter choice of a heuristic search are outside the literal-table census.

The direct files contribute 2,423 occurrences and 618 raw Git blobs. The incoming archives add 1,177 recognized occurrences. The combined inventory has 3,600 complete numeric binary-table occurrences in 667 byte forms. One four-state synthetic control graph, copied three times, has no separate halt target and is excluded from the one-halt recognizer. This leaves 3,597 occurrences and 666 byte forms. A six-state offset fixture and a nine-state bypass fixture have no root reaching all declared controls. The remaining 664 forms give 476 exact action-graph classes. None of these small fixtures is a claimed PA consistency machine.

Incoming archive	Recognized tables	194-state occurrences
N	215	23
R	256	59
Q	27	23
J/D	646	39
Radix	33	27
Total	1,177	171

Together with 90 direct-file occurrences, the 194-state portion has 261 occurrences, 51 byte forms and 28 graph classes. These counts include experimental inputs and explicitly rejected candidates; they do not mean 28 independently proved consistency machines.

Working states	Byte forms	Occurrences	Graph classes
6	1	5	0
9	1	3	0
194	51	261	28

Working states	Byte forms	Occurrences	Graph classes
195	26	198	16
196	163	1,198	99
197	27	67	15
198	12	37	5
199	10	34	4
200	4	14	2
201	8	28	7
202	15	36	15
203	5	12	4
206	2	6	1
207	58	175	57
208	177	1,134	142
209	20	123	13
210	19	57	19
211	14	45	11
212	16	50	14
213	17	46	9
214	5	15	4
215	2	8	2
217	2	6	1
218	2	8	1
219	3	9	3
220	2	6	2
221	2	6	1
222	2	10	1
Total	666	3,597	476

The complete raw corpus is retained in `comparison/raw-tables/` and `comparison/raw-tables-extra/`. Manifests map every occurrence to its pinned repository path or archive member. The full 41-packet contribution ledger is reproduced in Appendix B. Imported upstream snapshots and the repository's formalization modules supply dependencies and proof developments rather than additional novel packet designs.

2.2 How graph equivalence was tested

For every root reaching all working controls, and for each combination of global read/write symbol complementation and left/right reflection, we number controls by breadth-first traversal in read-bit order. We encode both outgoing actions of every control, distinguishing halt, and take the lexicographically least complete encoding. Equality of these encodings exactly decides the stated graph equivalence for this class of tables. The replay program compares complete strings; hash values are identifiers, not the basis of an assumed collision-free proof.

The comparison is stronger than checking the displayed start states or byte hashes. Both HA and HAZ are absent from all 28 prior classes. HAZ's canonical encoding has SHA-256

`a922e6d7cd9639ec9f7c6a3b953679133fc6625e5540daa532836a6d873d0478`.

This certifies structural difference from the recorded graphs. It is not a statement about semantic inequivalence: all accepted consistency machines are intended to have the same blank halting truth value.

2.3 The prior 194-state designs

Class or family	Mechanism and comparison
Primary PA194	Separate diagonal head, 67 raw AX readers and seven compatible folds: $134 + 67 - 7 = 194$.
PA194-AD	Same source; the physical locations of A and D are exchanged.
PA194-68	Alternative AX association: $134+68-8 = 194$, with the same resolved source.
C, altC	Cached-result MP and a cache-only Shift test. altC uses the alternative decoder. Several C packets repeat the same principal table.
CS, CZ, CSZ	CS uses symmetric destructive equality in Shift. CZ replaces a known-zero scheduler jump; CSZ combines both. R's named X machine is graph-equivalent to CSZ.
Q	Exchanges the premises of the sixth quantifier-distribution head; 79 resolved instructions change. It preserves the calculus through a propositional conversion.
J; J-A, J-H, J-X	Ten known-zero jump substitutions, with distinct final-register siblings. Source invariants are essential; a graph count alone cannot validate them.
D; D-CSZ-A	Generates $\Gamma(D)$ instead of $\Gamma(A)$, with adjusted selection. The primary D realization uses 69 readers and nine folds. The CSZ-based sibling is a distinct auxiliary.
M4R, M5, M6	A reassociated multiplier reader enables a fold. M4R preserves the original radix-four source; M5 and M6 change constructor coefficients. Each uses $135 + 67 - 8 = 194$.
N	Removes the successful-MP dummy push, returning the consequence in the visible top. It has 3,654 source sites and 4,755 outcomes.
MPH	Existing auxiliary in N. MP returns the consequence in the cache, while Shift retains $t + h = 1$. This is exactly the abandoned initial H candidate.
F, R, A	F uses a known-zero final AX decrement with symmetric Shift; R reassociates the emitter, changing 1,111 source addresses and 44 of 67 reader types. A instead replaces the scheduler entry jump by a known-zero decrement of A.
CS-both-zero6	Additional zero-test search input in R; retain its experimental status separately from principal outputs.
C-I-Z	Experimental cached-I zero-jump input in the radix archive.
Rejected scanner194	Confuses a legal seven-zero address with a separator; the reported failure is at source site 1003.
Rejected103	R archive's rejected candidate: source contracts pass, but blank initialization halts after 1,361 steps.
Rejected node1-X	Radix auxiliary explicitly rejected after a blank halt at step 1,361.

The preceding grouped rows contain exactly 28 graph classes. Individual names, complete digests and source members are in `comparison/graph-fixtures.json`; Appendix A lists short fingerprints for convenient identification. Historical source counts and acceptance claims above come from the supplied reports, whereas the literal-table census and candidate graph comparisons were newly executed.

The N and MPH distinction is substantive. For a successful MP with consequence b, primary PA194 returns $\langle b, [0; w], b, 0\rangle$, N returns $\langle b, w, b, 0\rangle$, and MPH/HA/HAZ return $\langle 0, w, 0, b\rangle$. Those

command interfaces require different proof compilers even though they search for refutations of the same calculus. C uses the last MP interface but changes the Shift halt test to cache-only.

2.4 Larger designs and why reductions do not simply add

The 195-state families include bundled-diagonal PA195 (3,700 source sites, 4,815 outcomes), separate-head 195S (3,673/4,780), cached 195C (3,649/4,749), cached bundled PA195DC (3,676/4,784), and the M195S8 scanner repair. The 196-state families include decoder association, register permutations, cached commands, known-zero jumps and logical predecessors of the 195/194 emitters. NEG18 shortens a source to 3,441 sites but costs 197 physical states. Gate simplifications and modal/explosion experiments include countermodels and failed candidates, not just successful reductions.

The early 208-state packets provide several proofs of the same native 172/211 helper fold. Later helper layouts introduce additional graphs. The imported PA209 lineage and U209 parents must be distinguished. The complete combined census records 142 classes at 208 states and 57 at 207, including search candidates; these are not counts of accepted theorems.

The seven-fold and eight-fold 194-state allocations cannot be added: their raw reader counts differ. Likewise, R/F's extra symmetric-controller state makes its allocation $135 + 67 - 8$, rather than the primary $134 + 67 - 7$. The new fold pays for the extra state. Radix M4R has the same accounting issue. Transplanting a named fold without its changed context or state budget does not produce a 193-state machine.

3 Numerical calculus and the hybrid command interface

Let

$$p(a,b) = \frac{(a+b)(a+b+1)}{2} + b, \qquad [a;w] = p(a,w).$$

The total inverse projections satisfy $\mathrm{hd}(0) = \mathrm{tl}(0) = 0$ and $[0;0] = 0$. A store is $\langle t, w, d, h\rangle$: visible theorem, encoded stack, numerical offset, cached theorem. A canonical store means $\langle t, w, 0, 0\rangle$; its top and stack are otherwise arbitrary.

Retain the numerical formula constructors

$$I(a,b) = 4p(a,b), \quad M(a,b) = 4p(a,b)+1, \quad U(a,B) = 4p(B,a)+2, \quad E(a,b) = 4p(a,b)+3.$$

Code zero is truth; code one is contradiction $M(0,0)$. Write

$$N(B) = I(B,1), \quad B \wedge C = N(I(B,N(C))), \quad \exists x_a B = N(U(a,N(B))).$$

The diagonal head is $\Gamma(B) = I(B,U(B,B))$.

The six ordinary heads are

$$\begin{aligned}
L_1(D,H) &= I(\exists x_D U(D,H), H),\\
L_2(D,A,H) &= I(U(D,U(A,H)), U(A,U(D,H))),\\
L_3(D,A,H) &= I(M(H,D), I(E(H,A), M(A,D))),\\
L_4(D,A,H) &= I(E(H,D), I(M(A,H), M(A,D))),\\
L_5(D,H) &= \exists x_D E(H,D),\\
L_6(D,A,H) &= I(U(D,A), I(U(D,I(A,H)), U(D,H))).
\end{aligned}$$

The dominated propositional head is

$$\rho(a,b,c,u) = I(I(I(c,u),a), I(I(a,c), I(b,c))), \qquad u \le a.$$

For the separation gate put

$$\begin{aligned} V(a,b,c) &= I(M(a,b), U(c, M(a,b))),\\ P(a,b,c) &= I(c, M(c,b)),\\ B(a,b,c) &= U\big(b, I(M(a,b), P(a,b,c)) \wedge I(M(c,b), c)\big),\\ Q(a,b,c) &= U(c, I(B(a,b,c), N(P(a,b,c)))). \end{aligned}$$

The guarded head $G(a,b,c)$ is zero if $c = a$ or $c = b$, and otherwise is $N(I(V(a,b,c), Q(a,b,c)))$. This is the unchanged numerical calculus of [3].

3.1 Commands

The hybrid Shift is

$$\mathsf{S}\langle t,w,d,h\rangle = \begin{cases} \mathsf{halt}, & t+h=1,\\ \langle h, \mathrm{tl}(w), d, \mathrm{hd}(w)\rangle, & t+h\neq 1. \end{cases}$$

For $y = d + x$, the changed MP command is

$$\mathsf{MP}_x\langle t,w,d,h\rangle = \begin{cases} \langle 0,w,0,y\rangle, & I(t,y)=h,\\ \langle 0,w,y,0\rangle, & I(t,y)\neq h. \end{cases}$$

Neither branch pushes on the stack. PA194C has this MP behavior but a different Shift test, $h = 1$.

For $\mathsf{AX}_{c,x,y}$ set $D = d + x$, $H = h + y$, $A = c$. AX returns top $G(H, D, A)$, clears offset and cache, and pushes the following ten entries, top first:

$$\Gamma(A), L_6, L_5, L_4, L_3, L_2, L_1, \rho(D, A, H, d), U(d, h), t.$$

The arguments of L_i are those displayed above. There are nine newly generated formulas and the saved incoming top.

Every generated formula except $U(d, h)$ is divisible by four. The universal formula is congruent to two modulo four and at least two. The gate is divisible by four. In particular, no generated formula is code one.

4 A total Pop operation and complete proof compilation

Throughout this section set $n = 10$, the exact push count of AX. Semicolon denotes execution in the written order. The total Pop identity also holds in the supplied uncached protocol; it is used here with the cached-result MP interface.

Lemma 4.1 (Total stack operations)**.** *Define*

$$\begin{aligned} \mathrm{Push}_n &= \mathsf{AX}_{0,0,0}; \mathsf{S}^{n-1}; \mathsf{MP}_0,\\ \mathrm{Pop}_n &= \mathsf{AX}_{0,0,0}; \mathsf{S}^{n+1},\\ \mathrm{Discard}_n &= \mathsf{MP}_1; \mathsf{AX}_{1,0,0}; \mathsf{S}^{n}; \mathsf{MP}_0. \end{aligned}$$

Then

$$\begin{aligned} \mathrm{Push}_n\langle t,w,d,h\rangle &= \langle 0, [t; w], 0, 0\rangle,\\ \mathrm{Pop}_n\langle t,w,d,h\rangle &= \langle t, \mathrm{tl}(w), 0, \mathrm{hd}(w)\rangle,\\ \mathrm{Discard}_n\langle t,w,0,0\rangle &= \langle 0, w, 0, 0\rangle. \end{aligned}$$

All three identities are total.

Proof. The intermediate generated formula codes cannot trigger the sum-one test. After $n-1$ Shifts in Push, the cache is $U(d,h)$, whereas every implication code is divisible by four; MP therefore fails at parameter zero and clears both visible registers, preserving $[t;w]$.

After n Shifts in Pop, the top is $U(d,h)$, the cache is the saved t, and the stack is the original w. The last Shift is safe because $U(d,h)+t \geq 2$. This handles $t=1$ explicitly.

In Discard, the first MP fails because $I(t,1) > 0$ and the cache is zero. It establishes offset one and zero visible theorems. After AX and n Shifts, top is $U(1,0)$, cache zero and stack w. The final comparison also fails since $I(U(1,0),0) > 0$. The proof uses pairing identities, not a false assumption that a zero push increases canonical stack length. □

Define Offset_0 to be the empty word and $\mathrm{Offset}_i = \mathsf{MP}_i$ for $i > 0$.

Lemma 4.2 (Generalization)**.** *The word*

$$\mathrm{Gen}_{i,n} = \mathrm{Push}_n; \mathrm{Offset}_i; \mathsf{S}; \mathsf{AX}_{i,0,0}; \mathsf{S}^n$$

maps $\langle a, w, 0, 0\rangle$ *to* $\langle U(i,a), w, 0, 0\rangle$ *for every* a, w.

Proof. Push stores a and clears the visible registers. Offset establishes $d = i$. Shift then loads a into the cache. The last generated AX head is $U(i,a)$, which the final n Shifts select while recovering the saved zero cache. No sum-one check fails, including when $a = 1$. □

For a head at top-down position $1 \leq j \leq n-3$, use

$$\mathrm{Select}_{j,D,H,A} = \mathrm{Discard}_n; \mathsf{AX}_{A,D,H}; \mathsf{S}^{j+1}; \mathrm{Pop}_n^{n-1-j}\,.$$

From any canonical incoming store $\langle t, w, 0, 0\rangle$ this selects that head, retaining w and zero offset and cache. At $n = 10$, positions $1, \ldots, 7$ are $\Gamma(A), L_6, L_5, L_4, L_3, L_2, L_1$. The dominated head with $u \leq a$ is selected by

$$\mathrm{Discard}_n; \mathrm{Offset}_u; \mathsf{AX}_{b,a-u,c}; \mathsf{S}^{n-1}; \mathrm{Pop}_n\,.$$

The separation gate is selected by

$$\mathrm{Discard}_n; \mathsf{AX}_{c,b,a}; \mathrm{Pop}_n^n\,.$$

These are the correctly adjusted ten-entry selectors. Old nine-entry indices would be wrong.

Lemma 4.3 (MP suffix)**.** *On* $\langle 0, [a, I(a,b), 0; w], 0, 0\rangle$*, the word* $\mathsf{S}; \mathsf{S}; \mathsf{MP}_b; \mathsf{S}$ *halts if* $a = 1$ *or* $b = 1$*; otherwise it returns* $\langle b, w, 0, 0\rangle$.

Proof. The successive nonhalting stores are

$$\begin{aligned}
&\langle 0, [I(a,b), 0; w], 0, a\rangle,\\
&\langle a, [0; w], 0, I(a,b)\rangle,\\
&\langle 0, [0; w], 0, b\rangle,\\
&\langle b, w, 0, 0\rangle.
\end{aligned}$$

The second Shift halts exactly when $a = 1$, and the last exactly when $b = 1$. The inserted zero restores the underlying stack. □

Proofs in the numerical calculus are built from the displayed axioms by MP and generalization. Define W_p by recursion on such a proof. An axiom uses its selector. Generalization at i uses $W_{p_0}; \mathrm{Gen}_{i,n}$. For MP with minor proof p_0 of a and major proof p_1 of $I(a,b)$, use

$$\begin{aligned}
W_p = {}& \mathrm{Discard}_n; \mathrm{Push}_n; W_{p_1}; \mathrm{Push}_n;\\
& W_{p_0}; \mathrm{Push}_n; \mathsf{S}; \mathsf{S}; \mathsf{MP}_b; \mathsf{S}.
\end{aligned}$$

Theorem 4.4 (All-store completeness)**.** *For every t, w, W_p from $\langle t, w, 0, 0\rangle$ either halts or returns $\langle \mathrm{concl}(p), w, 0, 0\rangle$. If p proves code one, then*

$$\mathsf{AX}_{0,0,0}; W_p; \mathsf{S}$$

halts from every incoming store.

Proof. Induct on p, using the selector, generalization and MP identities. In the MP case the first Discard and Push insert the needed dummy zero. An early halt is already a successful completeness outcome. For the last assertion, the initial AX clears offset and cache. If W_p returns, it leaves top one and cache zero, so the final Shift halts. □

This universal incoming-store property is the correct scheduler interface. The inherited scheduler carries its store between command words; a completeness proof assuming a fresh reset after every trial would be insufficient.

5 Soundness and PA formalizability

Annotate top, cache and every canonical live stack entry with a numerical-calculus derivation. The offset requires no proof. Code zero has a fixed derivation, obtained from the all-zero dominated propositional head.

AX supplies its axiom instances, generalizes the cached theorem to $U(d, h)$, preserves the incoming top and clears visible cache. Successful MP obtains a proof of y from proofs of t and $I(t, y)$ and puts y into the cache. Failed MP uses the fixed zero proof. A nonhalting Shift moves existing annotations.

If Shift halts, $t + h = 1$ implies $(t, h) = (1, 0)$ or $(0, 1)$. One of the visible annotations is therefore a refutation. Canonical stack expansion handles $[0; 0] = 0$: a pop of numerical zero supplies the fixed truth proof.

Theorem 5.1 (Command equivalence)**.** *The HA command enumerator halts exactly when its unchanged numerical calculus has a refutation.*

Proof. The annotation invariant gives soundness. The all-store contradiction word and the inherited exhaustive scheduler give completeness. □

The proof transformations used here are primitive recursive: pairing and projections, finite command words, recursive proof-tree compilation and annotation updates. Their correctness follows by induction on finite proofs, words and runs, all available in PA. No truth or reflection principle for PA is used.

The inherited numerical calibration [3] supplies proof translations between numerical refutations and PA refutations. Its ingredients include diagonal freshness, syntactic recovery of membership rewriting and equality interfaces, a finite-binary-membership interpretation, and uniformly proof-producing PA finite-comprehension induction templates. The exceptional code identity $I(0, 0) = 0$ is treated explicitly. This report retains that ordinary calibration; it does not replace it with a new kernel-checked endpoint.

The full inherited calibration texts are retained in `references/calibration/` and the pinned primary report sources in `references/primary_packet/report/`. In particular, `original_pa195_calibration.tex`, `diagonal_calculus.tex`, and `separate_head_calculus.tex` give the finite-binary-membership interpretation, the recovery of the removed equality-vacuity interfaces, and the bundled/separate-head translations. The command proof in this report supplies the appropriate hybrid interface; the cached-only proof is not substituted for it.

6 Source construction and physical verification

6.1 The HA source and register layout

The sole command-source edit, before the HAZ closure transformation, is

$$\texttt{node163} = (\texttt{node6}, \texttt{node147})$$

instead of $(\texttt{node6}, \texttt{node55})$ in the frozen program of [5]. The successful MP branch therefore uses the already supplied cache-return tail; it neither pushes a dummy zero nor returns the consequence as its visible top. Shift retains the original sum test. The AX grammar otherwise remains the primary grammar.

The logical-to-physical register map is changed from

$$(6, 2, 0, 4, 3, 7, 8, 5, 1) \quad \text{to} \quad (6, 2, 0, 3, 4, 7, 8, 5, 1),$$

in logical order $(N, T, W, A, D, H, X, Y, L)$. Only A and D exchange physical locations. The three placement edits in Section 1 implement that frame conjugacy. The symbolic checker is run with the changed map; correctness is not inferred merely from the apparent symmetry of A and D.

6.2 The four-site zero-register certificate

Let P be the resolved source program for the A/D-swapped cached-MP machine, and let P_Z replace only site 3553 of P:

$$P(3553) = \text{jump}(3637), \qquad P_Z(3553) = \text{dec}(6; 3637, 3554).$$

Here register 6 is X, and the first successor of a decrement is its zero branch. A zero branch leaves the tested register unchanged. Both programs have 3650 resolved source sites.

Put $Z = \{3550, 3551, 3552, 3553\}$. The following data are checked directly from the complete resolved instruction arrays, independently of the compiler and the symbolic Turing-machine simulator. The only edges into Z from outside Z are

$$3545 \xrightarrow{X=0} 3551, \qquad 3548 \xrightarrow{X=0} 3550.$$

Their source instructions are respectively $\text{dec}(6; 3551, 3546)$ and $\text{dec}(6; 3550, 3549)$. The unchanged instructions within Z are

$$P(3550) = \text{jump}(3553), \quad P(3551) = \text{dec}(1; 3553, 3552), \quad P(3552) = \text{jump}(3551).$$

In particular, no instruction within Z increments X.

The predicate

$$\mathcal{I}(pc, \mathbf{r}) \;\equiv\; (pc \in Z \implies r_6 = 0)$$

is therefore an inductive invariant of either source program. It is initially true at the source entry $pc = 25$, for arbitrary natural values of all nine registers. Every entry from outside Z establishes it by a zero test of X; every step from within Z preserves it or exits Z. At the sole changed site, $\mathcal{I}$ forces the new decrement to take its zero branch. That branch has the old jump's destination and makes no change to any register. Every other instruction is identical. Induction on source time now proves equality of the complete source configurations of P and P_Z at every step, until their common halt if one occurs. The extra positive branch in the resolved graph of P_Z is unreachable from the source entry.

This is an invariant argument for all register values and all execution lengths, rather than a bounded simulation result. It supplies the source equivalence needed to transfer the command-level soundness, completeness, and scheduler argument to the modified source. The independent blank-tape startup certificate and the universal instruction-refinement certificate remain necessary to connect that source theorem to the displayed binary Turing-machine table.

6.3 Universal source contracts and exact initialization

The HAZ grammar replaces the single occurrence

$$(\texttt{node150}, (\texttt{leaf}, \texttt{breakTo}, 1)) \quad \text{by} \quad (\texttt{node150}, (\texttt{leaf}, \texttt{dec}, 6)).$$

It changes the native row $(2066, 1)$ as specified above. Its extra positive source branch is included in the universal physical checks, even though the preceding invariant proves it unreachable from source entry 25.

Quantity	Primary PA194	PA194HA	PA194HAZ
Working controls	194	194	194
Transition rows	388	388	388
Primitive sites	3,673	3,650	3,650
Symbolic source outcomes	4,780	4,750	4,751
Verifier iterations	1,479,348	1,472,170	1,472,644
Maximum per outcome	497	497	497
Blank startup transitions	58,154	58,122	58,122

These are executed results. Python reproduces primary PA194 and its A/D layout before checking both new machines. The independently ported JavaScript resolver/runtime reproduces the A/D baseline and HAZ, then compares the complete tables, labels, source instructions and source addresses with the Python-produced files. Both implementations agree. The iteration counts measure symbolic-verifier work, not physical running time for arbitrary register values.

A contract represents arbitrary natural registers, the permitted arbitrary spare tail, and the affine header gap of the supplied runtime. Both branches of every decrement are checked. Complete tape frames and successor addresses are compared. The 23 arithmetic-box control rows are unchanged; their universal contracts are inherited. The runtime's oracle summarizes those blocks, so these runs do not independently establish their arithmetic correctness. The supplied `M6/Runtime.lean` context and its `BoxLemmas` assumptions, the primary report's block arguments, and the upstream references are preserved in the package.

The exact blank endpoint is native state 84 at head -2, source site 25, with logical registers

$$(N, T, W, A, D, H, X, Y, L) = (2, 0, 0, 0, 0, 0, 0, 0, 0),$$

and 26 spare blocks of value one. The complete finite-support tape matches the expected frame, including the zero exterior. It is not a window-only or register-only startup test. The full list of one-cells appears in both execution receipts.

Each nonhalting primitive source step has a positive-duration physical realization with its complete frame and correct successor. Concatenation gives physical completeness. For the converse, suppose the physical run first halts at time T. If the source did not halt, $T + 1$ nonhalting source steps would give a nonhalting physical run beyond time T, contradicting determinism. This finite argument is formalizable in PA and avoids any assumption that a long observed run proves nonhalting.

Theorem 6.1 (PA calibration of the literal tables)**.** *Using the unchanged inherited arithmetic-block and numerical-calibration lemmas specified above, for* $M \in \{\text{PA194HA}, \text{PA194HAZ}\}$*,*

$$\boxed{\text{PA} \vdash (\neg \text{Halt}(M, \textit{blank}) \leftrightarrow \text{Con}(\text{PA})).}$$

Proof. The HA command compiler, soundness invariant and exhaustive scheduler translate between finite source halts and numerical refutations. Compose with the inherited PA-refutation

translations. The four-site invariant transfers this source theorem to HAZ. Compose each source theorem with its universal positive-duration refinement and exact initialization. Negate the resulting equivalence of existential halting and PA-refutation predicates. All new finite transformations and inductions are available in PA; no PA reflection principle is used. □

The receipts explicitly record `kernel_checked: false`. This qualification concerns the form of verification, not an assumption of Con(PA) in the equivalence theorem.

7 Attempts to get below 194 states

7.1 Bounded physical quotients

Three sufficient contraction rules were implemented. The first merges a control into another through a bounded local detour that restores the complete local tape, head and endpoint modulo the proposed alias. The search includes arithmetic controls. The second represents the removed control with a head displacement of ± 2, compensating incoming movement. The third represents it by toggling a neighboring marker bit, compensates the incoming write, and requires restoration before continuation.

A radius-r abstract context retains the $2r + 1$ cells around the head. Boundary transitions admit both possible new bits, giving a conservative reachable-window invariant. The limits below restrict the sufficient certificate rules and detour lengths, not the represented natural register values. Failure to find such a certificate is not a lower bound.

HAZ bound	Local contexts	Translated cases	Marker cases	Reductions
Radius 6, 256 steps	274,499	32,560	32,366	0
Radius 8, 384 steps	3,181,913	32,560	32,366	0

At radius eight, the translated search examined 1,525,492 actions and the marker search 941,501. Both HA runs were also negative. Earlier recorded runs cover primary PA194, AD, MPH, Q, J, D, M4R, M5, M6 and CSZ. A final radius-six literal-head pass covered the five newly catalogued nonrejected auxiliary 194-state graphs. No validated 193-state result emerged.

A separate limited AX-grammar exploration evaluated 22,681 proposals while preserving the primary 190-token emitter. It retained effective reader score 60 and found no score 59. Targeted attempts to combine the final AX zero decrement with the R/F controller fold likewise found no saving. All these results are scoped negative experiments. The archive includes the exact search implementations and numerical receipts, so their scope can be inspected rather than inferred from the number of trials.

7.2 Transfer from ZFC and concise iPRA

The ZFC omit-zero alternative [7] shows that removing the successful-MP dummy push can preserve the consistency target after changing the proof compiler. It changes native target 98 to 114, removing 19 source primitive occurrences while retaining 218 physical states in that ZFC construction. PA194N is the corresponding supplied PA no-push direction. HAZ instead uses target 130 and the cached-result interface. Source trace equality is not available for these MP changes, which is why the explicit command proof is needed.

Concise iPRA [8] suggests restricted term forms, checking a rule in its consumer context, and syntactic proof translations. Its scalar core uses zero, successor and iteration; its equality, Q and D formula forms are useful checker-compression ingredients. A PA-calibrated refutation checker built

from such a core remains a possible direction. The existing iPRA checker alone has a different consistency target. Its short high-level source also does not price the binary-tape register engine, scheduler and blank initialization needed for a state bound.

The most direct remaining physical target is an effective AX-reader score 59 with the 134-state fixed core, or a certified net saving in that core. A useful next search would jointly vary the emitter grammar and a proved tape marker invariant, rather than count nominal folds independently of the states that implement their contexts. These are research proposals, not unverified machines offered as solutions.

8 Reproduction and contents

The delivered package contains all inputs needed for the new-machine checks and corpus replay without a network connection. From its root, run the following commands with Python assertions enabled:

```
python3 build.py
node verification/verify_haz.cjs
python3 verification/haz_four_site_cut.py \
  machine/pa194ha.instructions.json \
  machine/pa194haz.instructions.json
node comparison/verify-corpus.js
python3 comparison/check-census.py
node comparison/check-combined-corpus.js
```

The Python and JavaScript construction checks, independent invariant checker, and corpus checks were executed successfully during this work. The optional larger graph-census replay recomputes all 476 classes; the small `verify-corpus.js` check handles the 28 prior 194-state classes and both candidates.

The bounded HAZ search can be repeated with

```
node search/search.cjs machine/pa194haz.native.tm 74 8 384
```

For the report, run `latexmk -pdf pa194haz.tex` in `report/`, or run `pdflatex -halt-on-error pa194haz.tex` twice. This PDF was compiled from the supplied LaTeX and visually inspected. Python 3.10 or newer, a recent Node.js with standard modules, and a conventional TeX Live installation suffice. No Lean installation is required for these finite checks; compiling a new kernel proof is not one of the reproduced steps.

The main deliverables are the native/dense machines, label maps, complete resolved sources, manuscript and PDF. The package also contains execution receipts, the independent invariant certificate, the 41-packet ledger, all 667 prior raw byte forms, graph representatives and manifests, search code/results, inherited proof texts, provenance and licenses. A top-level `SHA256SUMS` authenticates the packaged files.

The calculus, register engine, compilation interfaces and original 194-state allocation are credited to the supplied ProveIt work and James Chang's imported PA196 development. HA and HAZ are new concrete compositions relative to the fully inspected literal corpus. Their mathematical and physical justification is made explicit here. The comparison also records failed candidates so that an appealing but unsound smaller table is not silently counted as an improvement.

References

[1] ProveIt, PA research status. Pinned PA README.

[2] ProveIt, primary PA194 construction, register-layout companion, and physical proof. Pinned primary packet.

[3] ProveIt, primary PA194 numerical-calculus calibration and references. Pinned calibration source.

[4] ProveIt, cached command selectors and their PA calibration. Pinned cached-protocol packet.

[5] James Chang / ProveIt, imported PA196 source program. Pinned Program.lean.

[6] ProveIt, five incoming N, R, Q, J/D, and radix archives. Pinned incoming directory; individual archive and member identities are in the packaged manifests.

[7] ProveIt, ZFC omit-successful-MP-zero alternative. Pinned omit-zero README.

[8] ProveIt, Concise iPRA project. Pinned ConciseIPRA README.

A The 28 prior 194-state graph classes

The following prefixes identify the complete canonical graph digests in `comparison/graph-fixtures.json`. These are graph identities, which usually differ from native and dense file digests. The candidate files are kept separately. Several normalized comparison tables have starts 67 or 68; their JSON start field is authoritative.

Class	Canonical SHA-256 prefix
PA194	ac035e6add185fa82fc7
PA194_AD	7f4865bbf7a51d30d13e
PA194_68	9fcc36cb1f5e4ae7c51c
PA194C	a852f4c11d61a4a323a6
PA194CS	9907ae28ce2c16c676db
PA194CZ	4448c7f90278d867014d
PA194altC	2fd5606a2ddc95432255
rejected_scanner194	872c6c0c18d62460b237
Q	2b7eb597816c6aa8f458
CSZ	f9a25f6cce889a23bf94
J	36d2949a4beec8bb100a
D	d98a45b651b5dd745b1d
M4R	7586918e1fc564279f27
M5	445931f0eef43940d8ea
M6	e352bc7d2b60c2043ec7
N	38ecf5d7b2937e0edb50
MPH	505a918530a22509b2bd
PA194A	4ff967da98e145109790
PA194F	ec72d8c7b3ceb29c6ca7
PA194R	946e9a9cf2b2653d8ef7
rejected103	2230f5a83b8a53072c4b
CS_both_zero6	cdc8184eda97708ca858
D_CSZ_A	6ee92c7bd3ee82027e10

Class	Canonical SHA-256 prefix
J_A	7e3aef291f3783608d63
J_H	8fedfc6e44a68ec700c2
J_X	c9fe5e510e9ec12adbda
C_I_Z	32f8caeca7516d39eb15
rejected_node1_X	3a494eb08dc9f6b6e084

B All 41 placed research packets

The names below are relative to Computability/BusyBeaver/Research/PA/docs/ at the pinned revision. Summaries are report-derived. Every packet's direct table occurrences are included in the census even where its main contribution is a proof or a failed search. Full URLs, Git trees and counts are retained in comparison/packet-ledger.json.

No.	Packet and contribution
1	Below_U208_Research_2026-10-29 Guarded source pruning from 3909 to 3904 sites; no supported sub-208 physical result.
2	ConPA_below208_research_2026-10-29 Role-collapse experiments, a countermodel, and enumeration of 279936 assignments; no accepted reduction.
3	PA195_ConPA_research_2026-11-14 Separate-head PA195S with 3673 sites and cached-controller PA195C with 3649 sites.
4	PA_consistency_TM_research_2026-11-13 Five-zero address geometry and a 157-node decoder study.
5	U208_ConPA_2026-10-29 208-state realization identifying native controls 172 and 211; alternative presentations of the same saving.
6	U208_ConPA_Conditional_Context_2026-10-29 The common 208-state construction with a conditional-context proof of its native helper fold.
7	U208_ConPA_Entry_Context_2026-10-29 The common 208-state construction with entry-context justification of the same helper sharing.
8	U208_ConPA_History_2026-10-29 The common 208-state construction with history-based justification of the same saving.
9	conpa_tm_comparison_and_reduction_search 196-state modus-ponens variants; rejected Shift edits include a table that halts from blank after 43563 steps.
10	conpa_tm_comparison_and_search Accepted A/D physical-register exchange at 196 states; a two-rotation 195-state attempt fails at source site 93.
11	conpa_tm_study_2026-11-13 Combined 196-state controller: 3788 sites, 4928 outcomes and reported startup 57922; a direct-negation alternative costs 197 states.
12	pa194C_conpa_2026-11-13_2 PA194C caches successful MP results and uses a cached-head Shift; 3649 sites and 4749 outcomes.
13	pa194_conpa_2026-11-13 Primary PA194 uses 134+67−7 states; the same source also has an A/D-register-location variant.
14	pa194_conpa_2026-11-13_2 Alternative allocation 134+68−8 realizes the same 3673-site source at 194 states.

No.	Packet and contribution
15	`pa194_new_designs_2026-11-13` Repeats PA194C and adds PA194CS, whose symmetric destructive Shift equality reduces the source to 3646 sites.
16	`pa194c_conpa` Repeats PA194C and retains the alternative-decoder cached auxiliary PA194altC.
17	`pa194c_conpa_2026-11-13` Repeats the same PA194C literal graph with additional searches.
18	`pa194c_conpa_2026-11-13_1` Repeats PA194C and adds PA194CZ, replacing the scheduler jump by a decrement of known-zero X.
19	`pa195_conpa_2026-11-13` Original 195-state realization: 3700 source sites and 4815 outcomes.
20	`pa195_diagonal_combined_2026-11-13` PA195DC combines the diagonal and cached controller: 3676 sites and 4784 outcomes; positive and negative diagonal auxiliaries are included.
21	`pa195_followup_research` A nominal 194-state scanner fails on a legal address; a 195-state repair is supplied.
22	`pa196_research_2026-11-13` 196-state decoder with 157 nodes; the missing helper needed by a further reduction is identified.
23	`pa208_research_2026-10-29` Additional helper-sharing presentations; its report groups 27 presentations into 20 action-graph classes.
24	`pa_below196_research` Gate, orientation, and freshness searches without an accepted sub-196 result.
25	`pa_consistency_comparison_2026_11_13` Modal/explosion splitting, Hilbert certificates, and countermodels delimit valid logical simplifications.
26	`pa_consistency_comparison_and_new196_2026-11-13` Bundled and separate-diagonal logical parents with 196-state physical implementations.
27	`pa_consistency_cross_archive` Cross-comparison of diagonal, separate-head, Combined, and optional zero-jump 196-state variants.
28	`pa_consistency_reduction` Rebracketed and cached-head Shift variants at 196 states, plus larger gate alternatives and failed 195-state attempts.
29	`pa_consistency_reduction_audit_2026-11-14` Reduction audit with no newly accepted smaller table.
30	`pa_consistency_research` Decoder reassociation needs an added scanner, retaining 196 states; the nominal 195-state implementation fails.
31	`pa_consistency_research_2026-10-29` Guarded source pruning without a corresponding physical state saving.
32	`pa_consistency_research_2026-11-13` Diagonal-vacuity axiom simplification: 201 to 191 macro tokens and 3812 to 3700 primitive sites; initial physical realization has 199 states.
33	`pa_consistency_research_2026-11-13_2` Rotated 196-state decoder, 198-state diagonal variant, and an MP-halting source proposal without a smaller physical realization.
34	`pa_consistency_research_2026-11-14` NEG18 shortens a 3673-site source to 3441 sites but raises the physical count to 197 states.
35	`pa_tm_comparison_and_reductions` Rotated 196-state decoder and a 199-token, 3773-site gate whose physical realization has 198 states.

No.	Packet and contribution
36	`pa_tm_extended_research_2026-11-13` Stronger register-zero and difference invariants; four closing variants are alternative presentations of the same physical transition.
37	`pa_tm_research_2026-11-13` 196-state decoder work shortens maximal source address paths from 38 to 36.
38	`pa_tm_research_2026-11-13_2` Known-zero jump realization at 196 states: 4959 outcomes and reported startup 58402; an attempted helper deletion chooses a wrong register.
39	`pa_unary_power_cover` Logical calibration through unary power cover; no smaller literal Turing machine is supplied.
40	`u208_comparison_research_2026-10-29` Comparison of the original 208-state classes; a proposed 207-state table is rejected.
41	`u208_research_2026-10-29` The 208-state helper fold with symbolic source contracts and contextual checks.

C Complete HAZ transition table

An entry wDq writes w, moves in direction D, and enters q. Start is zero; working states are $0, \ldots, 193$; halt is 194.

State	Read 0	Read 1
0	1R1	1R2
1	1R3	1R1
2	0R0	1R2
3	0R4	0R1
4	1R5	1L6
5	1L7	1L8
6	0L7	1L6
7	0L9	1L6
8	0L10	0L11
9	1R12	0L9
10	0L9	0L10
11	1L8	1L11
12	0R12	0R13
13	0L14	0R13
14	1L15	0L15
15	0L15	1L16
16	0L17	1L16
17	0L18	1L16
18	0L19	1L16
19	0L20	1L16
20	0L21	1L16
21	0L22	1L16
22	0L23	1L16
23	0R23	1R24
24	0R25	1R26
25	0R27	1R28
26	0R29	1R30
27	0R31	1R194
28	0L32	1R33
29	0R34	1R35
30	0L36	1L37

State	Read 0	Read 1
31	0R38	1R39
32	0R40	1R41
33	0R42	1R43
34	0R44	1R45
35	0R46	1R47
36	0R48	1R49
37	0R50	1R51
38	0R52	1R42
39	0R53	1R54
40	0R55	1R56
41	0R57	1R58
42	0R59	0R60
43	0R61	0L14
44	0R62	1R63
45	0R55	1R64
46	0L65	1R66
47	0R36	1R67
48	0R59	0R68
49	0R69	0L14
50	0R70	0R71
51	0R72	0L49
52	0R73	1R54
53	0L74	0L15
54	0R75	0L10
55	0R76	1R77
56	0R78	1R79
57	0R80	1R81
58	0R82	1R83
59	0L84	0R1
60	0R85	1R60
61	0L76	1L86
62	0R87	0L7
63	0R88	1R89
64	0R42	1R90
65	0R91	1R49
66	0R92	0L75
67	0R86	1R33
68	0R93	1R68
69	0L94	1L33
70	0L84	0R2
71	0R95	1R71
72	0R75	1R96
73	0R59	0R85
74	0R86	1R97
75	0R42	0L13
76	0R87	1R37
77	0R30	1R98
78	0R42	1R99
79	0R100	1R101
80	0R102	1L103
81	0R30	1R80
82	0R97	1R104
83	0R30	1R105
84	1R84	1L15
85	0R106	1R85
86	0R73	1R107
87	0L46	0R93

State	Read 0	Read 1
88	0R108	1R98
89	0R108	1R69
90	0R109	0L14
91	0R110	1L74
92	0R111	1R112
93	0R113	1R93
94	0R50	1R103
95	0R114	1R95
96	0R30	1R49
97	0R48	1R61
98	0R87	1L115
99	0R104	0L14
100	0R36	1R116
101	0R115	1R78
102	0L46	0R113
103	0L45	1R117
104	0L94	1R118
105	0R119	0L14
106	0R71	1R106
107	0R120	1R30
108	0L46	0R106
109	0R70	0R114
110	0R59	0R114
111	0L65	1R121
112	0R122	1R123
113	0R14	1R113
114	0R68	1R114
115	0R124	1L67
116	0R125	1R126
117	0R120	0L14
118	0R94	1L98
119	0R127	1R127
120	0R70	0R68
121	0R128	1R129
122	0R32	1R36
123	0R31	1R115
124	0R59	0R95
125	0R102	1R117
126	0R82	1R81
127	0R120	1R120
128	0R130	1R131
129	0R132	1R133
130	0R134	1R135
131	0R136	1R137
132	0R138	1R139
133	0R140	1R141
134	0R100	1R67
135	0R142	1R127
136	0R143	1R144
137	0R145	1R146
138	0R134	1R147
139	0R120	1R148
140	0R149	1L131
141	0R150	1R151
142	0R79	1R41
143	0R28	1R152
144	0L153	1L154

State	Read 0	Read 1
145	0L129	1R155
146	0R156	1R157
147	0R149	1R158
148	0R159	1R120
149	0R100	1R160
150	0R161	1R162
151	0R163	1R42
152	0R164	1R165
153	0R166	1R157
154	0R167	1R156
155	0R135	1R130
156	0R147	1R41
157	0R134	1R168
158	0R41	1L169
159	0L170	1R171
160	0R91	1R64
161	0R172	1R170
162	0R173	1R174
163	0R175	1R176
164	0R177	1R178
165	0L179	1L180
166	0R149	1R140
167	0R165	1R181
168	0L130	1R120
169	0R182	1R135
170	0R183	1R145
171	0R158	1R130
172	0R139	1R165
173	0R158	1R184
174	0R185	1R91
175	0R31	1R53
176	0R186	1R175
177	0R134	1R142
178	0R32	1R187
179	0R144	1R188
180	0R189	1R190
181	0R41	1R191
182	0R168	1R41
183	0R192	1R120
184	0R153	1R146
185	0R165	1R180
186	0R96	1R115
187	0R193	1R159
188	0R169	1R154
189	0R158	1R192
190	0R179	1R182
191	0R182	1R41
192	0R149	1R168
193	0R79	1R144